

Biometric-Based Swimming Pace Prediction Using Machine Learning

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1 Introduction and Motivation

Swimming performance analysis has become an important area in sports analytics and machine learning research. Coaches and athletes continuously seek methods to improve training efficiency, predict race outcomes, and optimize athletic performance.

Traditional sports analysis methods are often based on manual observation and historical statistics. However, machine learning allows researchers to identify hidden patterns and relationships within athlete data that may not be visible through conventional analysis.

This project focuses on predicting swimming pace per 100 meters using biometric and competition-related features. By analyzing swimmer characteristics such as age, height, weight, sex, nationality, and race distance, machine learning algorithms can estimate swimming pace and performance outcomes.

The main goal of this project is to develop a regression-based machine learning system capable of predicting swimming pace using real-world swimming data collected from Olympic competition sources.

The project demonstrates how machine learning techniques can support sports analytics, athlete evaluation, and performance optimization.

2 Dataset Collection and Web Scraping

The dataset used in this project was collected from official Olympic Games Paris 2024 websites. Data collection was performed using web scraping techniques implemented in Python.

The following official Olympic resources were used:

- Olympic athletes database
- Official swimming competition results

Athlete-related information such as swimmer names, nationality, and participation details was collected from the athlete database. Swimming results including swimming distance and race performance were collected from the official swimming results website.

The collected data included:

- Swimmer name
- Nationality
- Swimming distance
- Swimming performance results
- Competition information

After collecting competition-related information, swimmer biometric data such as height, weight, age, and sex were associated with athletes using swimmer names as identifiers.

Web scraping was implemented using Python libraries including:

- Requests
- BeautifulSoup
- Pandas

The overall data collection workflow consisted of the following steps:

1. Collecting athlete information from Olympic athlete pages
2. Collecting official swimming race results
3. Matching swimmer names with biometric data
4. Cleaning and preprocessing collected data
5. Creating the final machine learning dataset

This approach allowed the creation of a structured swimming dataset containing both biometric and race-related information suitable for regression analysis.

3 Dataset and Data Preparation

The final dataset contains biometric and race-related information for swimmers participating in Olympic swimming competitions.

The dataset includes the following features:

- Sex
- Nationality
- Height

- Weight
- Age
- Swimming distance
- Simulated pace per 100 meters
- Simulated race time

The dataset contains both categorical and numerical variables, making it suitable for machine learning preprocessing and regression analysis.

3.1 Data Cleaning

Several preprocessing steps were performed before model training:

- Duplicate rows were removed
- Missing values were filled using statistical methods
- Swimmer IDs and names were removed from training features
- Numerical and categorical preprocessing was applied

For numerical columns, missing values were replaced using median values. For categorical variables, missing values were replaced using mode values.

3.2 Feature Engineering

Additional features were created to improve model performance.

3.2.1 BMI Calculation

Body Mass Index (BMI) was calculated using the following formula:

$$BMI = \frac{weight}{height^2}$$

where height is represented in meters.

BMI provides additional information about swimmer body composition and physical characteristics.

3.2.2 Distance Category

Swimming distances were categorized into three groups:

- Sprint
- Middle Distance
- Long Distance

Feature engineering improved the ability of machine learning models to identify relationships between swimmer characteristics and swimming pace.

4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to analyze patterns and relationships within the dataset.

Several visualizations were created during the analysis process:

- Distribution of swimming pace
- Height versus swimming pace
- Weight versus swimming pace
- Swimming pace grouped by sex
- Swimming pace grouped by distance category
- Correlation heatmap

The analysis demonstrated that swimming distance and biometric characteristics influence swimming pace prediction.

Correlation analysis showed relationships between physical body measurements and swimming performance indicators.

The target variable used in this project was:

`sim_pace_sec_per_100m`

This variable represents swimming pace measured in seconds per 100 meters.

5 Machine Learning Experiments

The project applies supervised machine learning regression techniques for swimming pace prediction.

5.1 Train-Test Split

The dataset was divided into:

- 80% training data
- 20% testing data

The split was performed using a random state value of 42 to ensure reproducibility.

5.2 Preprocessing Pipeline

The preprocessing pipeline included:

- StandardScaler for numerical feature normalization
- One-hot encoding for categorical variables

The preprocessing pipeline ensured that machine learning models could effectively process mixed data types.

5.3 Machine Learning Models

Five machine learning regression models were trained and evaluated:

1. Linear Regression
2. Decision Tree Regressor
3. Random Forest Regressor
4. Gradient Boosting Regressor
5. XGBoost Regressor

5.4 Evaluation Metrics

The following evaluation metrics were used to assess model performance:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R-squared Score (R^2)

5.4.1 Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

5.4.2 Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

5.4.3 R-squared Score

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

6 Hyperparameter Tuning

Hyperparameter tuning was applied to the Random Forest Regressor to improve prediction performance.

The following parameters were optimized:

- Number of estimators
- Maximum depth
- Minimum samples split
- Minimum samples leaf

GridSearchCV and manual parameter tuning approaches were used to identify the best-performing model configuration.

7 Results and Model Comparison

All machine learning models successfully predicted swimming pace using biometric and competition-related features.

The results demonstrated that ensemble learning algorithms such as Random Forest and XGBoost achieved better prediction performance compared to traditional linear models.

The best-performing model was selected using the following criteria:

- Lowest Mean Absolute Error
- Lowest Root Mean Squared Error
- Highest R^2 score

Feature importance analysis revealed that swimming distance, BMI, height, weight, and age strongly influenced swimming pace prediction.

The results demonstrate that biometric data can effectively support sports performance prediction using machine learning algorithms.

8 Error Analysis

Prediction errors were analyzed using multiple visualization and statistical approaches.

The following methods were applied:

- Actual versus predicted value plots
- Error distribution analysis
- Absolute error analysis

The majority of prediction results were close to actual swimming pace values, indicating strong regression performance.

However, some extreme performance cases produced larger prediction errors.

Potential limitations include:

- Limited environmental variables
- Simulated performance assumptions
- Missing physiological measurements
- Lack of training intensity information

Despite these limitations, the model demonstrated effective predictive capability.

9 Interactive Demonstration Application

An interactive prediction application was developed using Gradio and Streamlit technologies.

The application allows users to input swimmer information including:

- Sex
- Nationality
- Height
- Weight
- Age
- Swimming distance

The trained machine learning model predicts:

- Swimming pace per 100 meters
- Estimated total race time

The interactive application demonstrates the practical usability of the developed machine learning system and allows real-time prediction generation.

10 Related Research

Several studies related to sports analytics, swimming performance prediction, and machine learning applications in athletics were reviewed during the research process.

Existing research demonstrates that machine learning algorithms can effectively model athletic performance using biometric and physiological data.

Regression algorithms, ensemble learning methods, and neural networks are commonly used in sports prediction systems.

The reviewed literature supports the application of machine learning methods for predictive sports analytics and confirms the relevance of this project.

11 Conclusion

This project developed a machine learning system capable of predicting swimming pace using biometric and race-related data collected from Olympic competition sources.

The project successfully implemented:

- Web scraping and dataset collection
- Data cleaning and preprocessing
- Feature engineering
- Exploratory Data Analysis

- Multiple machine learning models
- Hyperparameter tuning
- Error analysis
- Interactive prediction application

The results demonstrate that machine learning algorithms can effectively predict swimming pace using swimmer biometric characteristics.

Ensemble learning models such as Random Forest and XGBoost achieved strong predictive performance and outperformed traditional regression methods.

The developed system has potential applications in sports analytics, athlete evaluation, training optimization, and performance forecasting.

Future work may include:

- Using larger real-world swimmer datasets
- Integrating wearable sensor data
- Including physiological measurements
- Applying deep learning techniques
- Expanding the dataset with additional competition variables

12 References

1. International Olympic Committee. Paris 2024 Olympic Games Athletes Database. Available at: <https://www.olympics.com/ru/olympic-games/paris-2024/athletes>
2. International Olympic Committee. Paris 2024 Swimming Results Database. Available at: <https://www.olympics.com/ru/olympic-games/paris-2024/results/swimming>
3. Géron, A. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow. O'Reilly Media, 2022.
4. Pedregosa, F., et al. Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 2011.
5. Chen, T., & Guestrin, C. XGBoost: A Scalable Tree Boosting System. Proceedings of the ACM SIGKDD Conference, 2016.
6. James, G., Witten, D., Hastie, T., & Tibshirani, R. An Introduction to Statistical Learning. Springer, 2021.
7. Bishop, C. Pattern Recognition and Machine Learning. Springer, 2006.